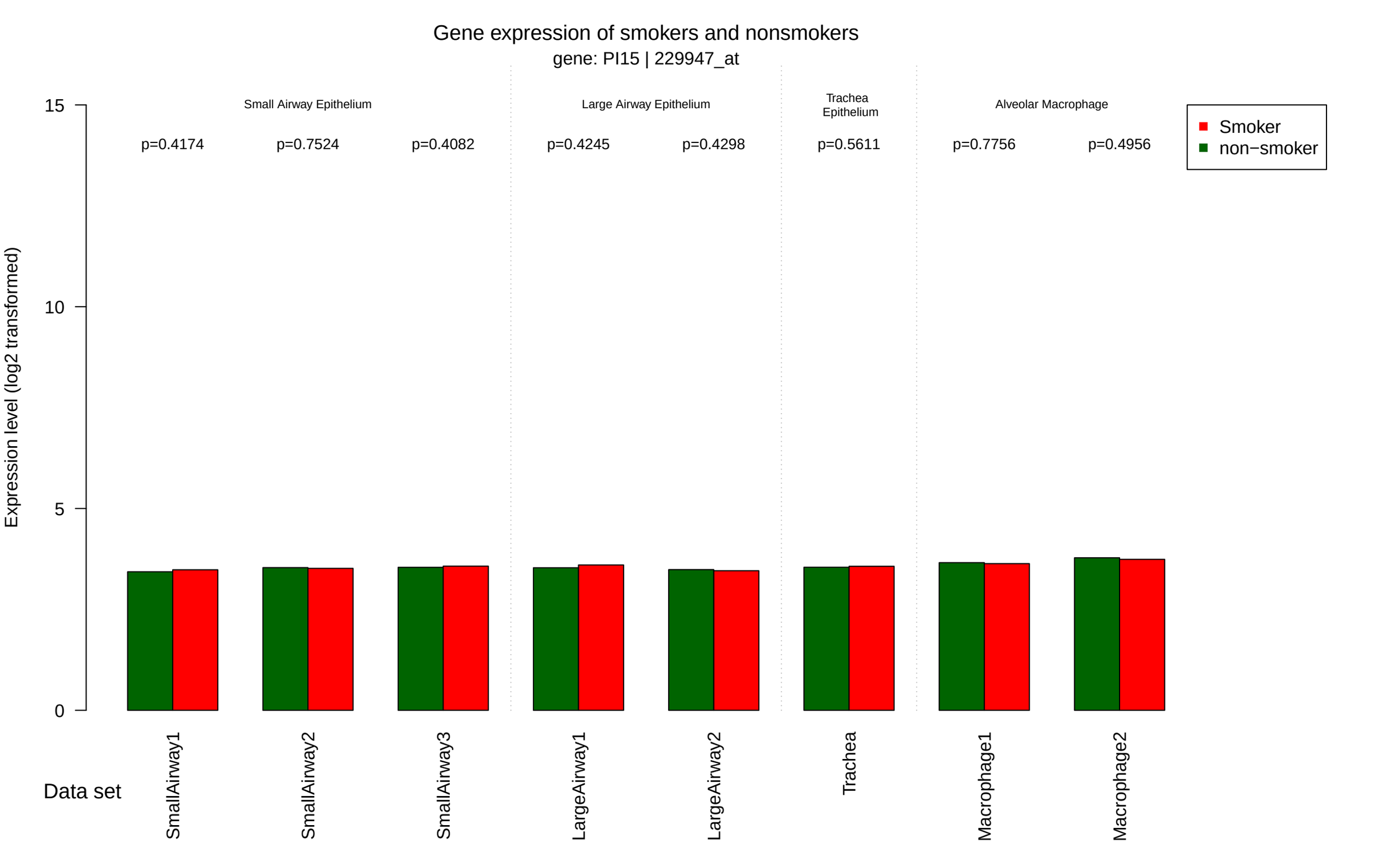
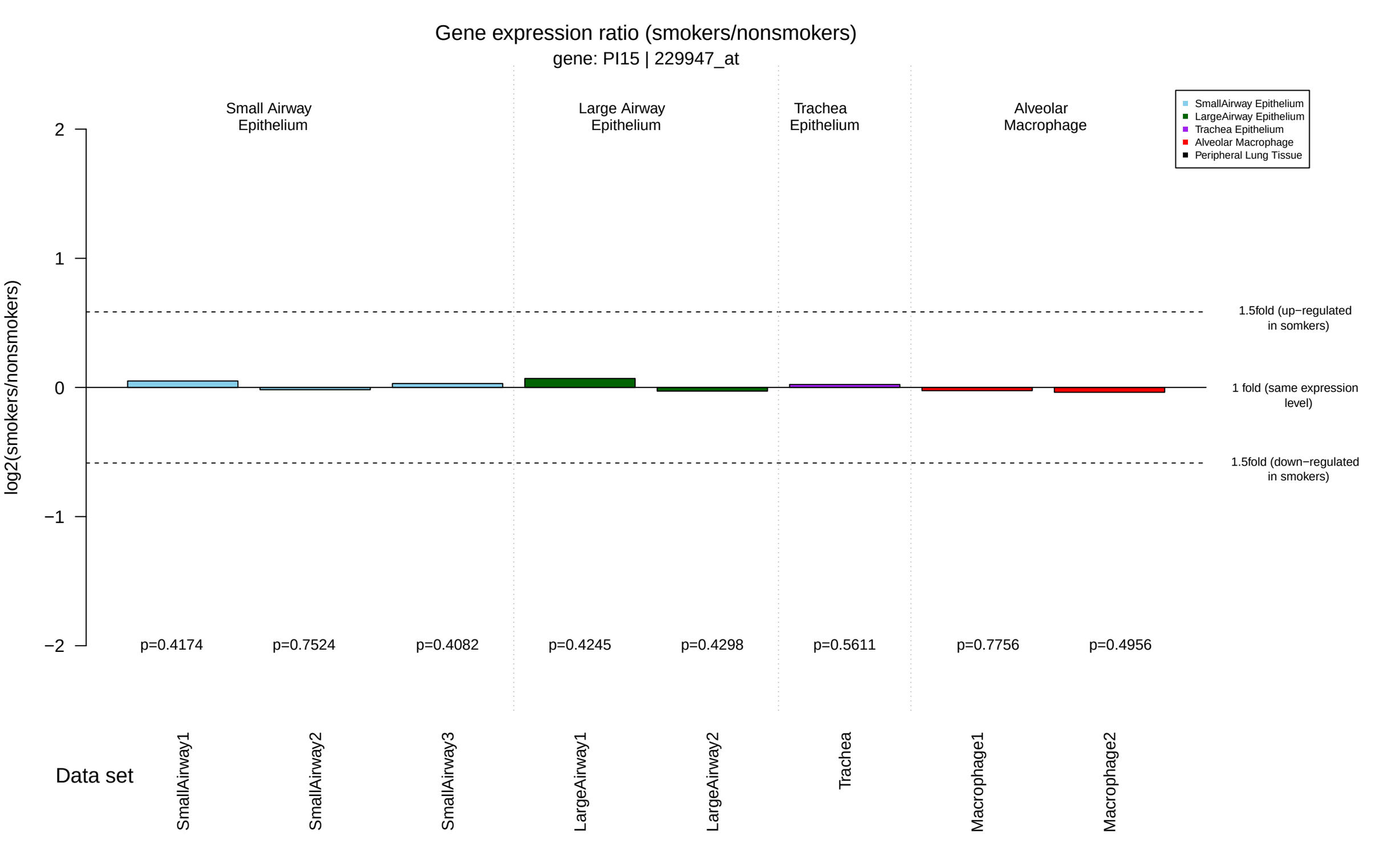
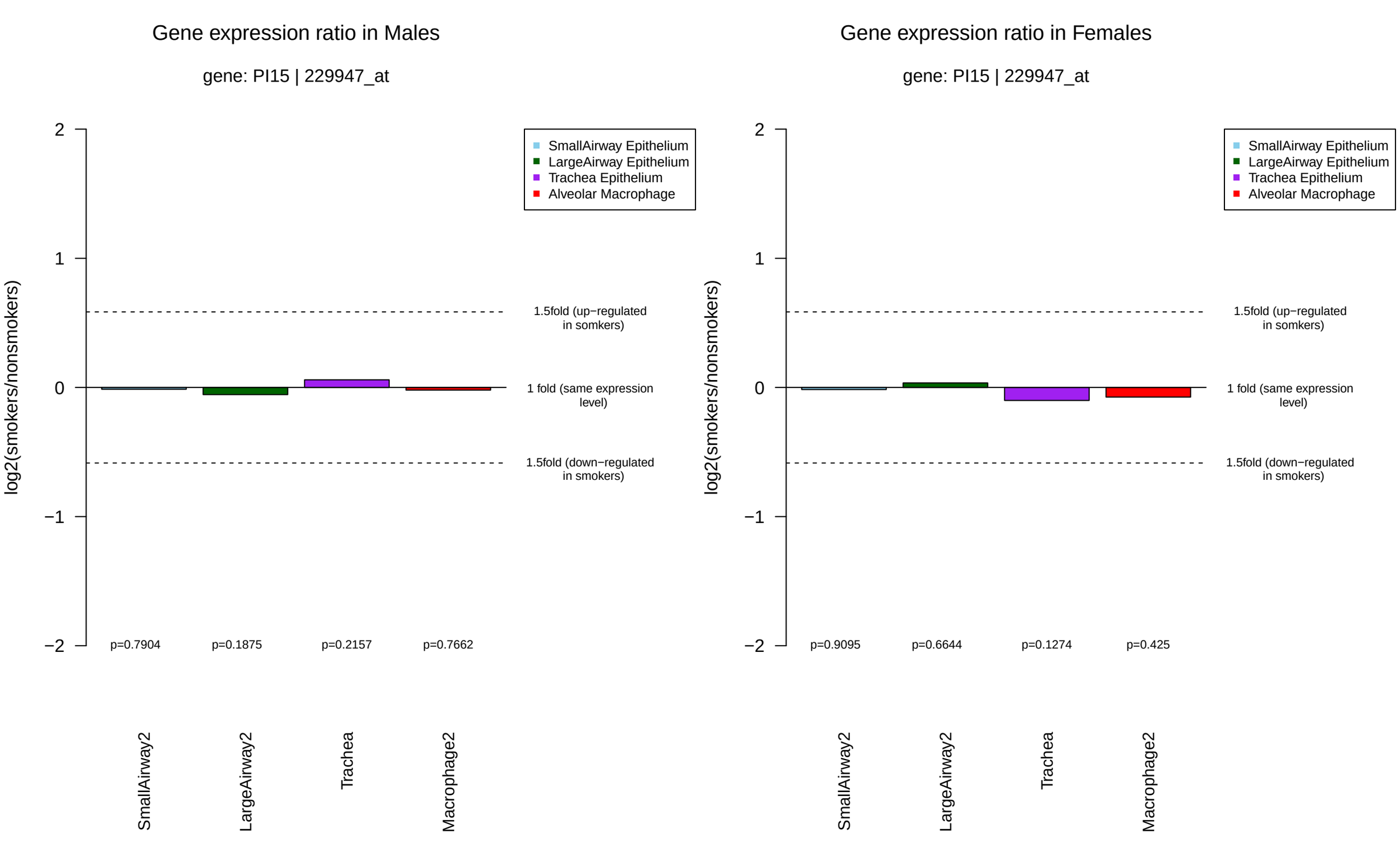
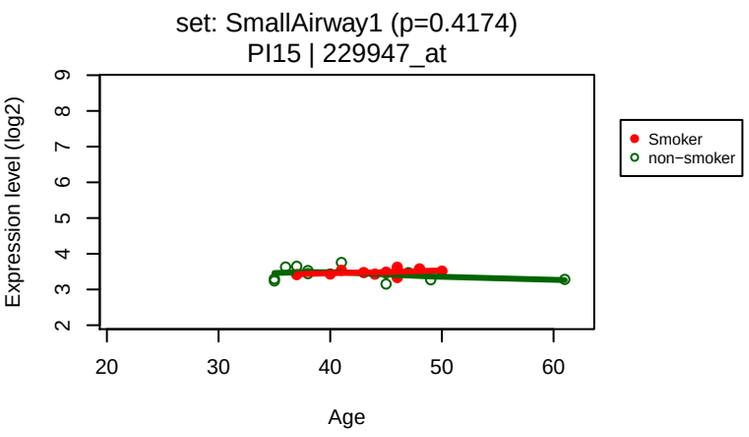
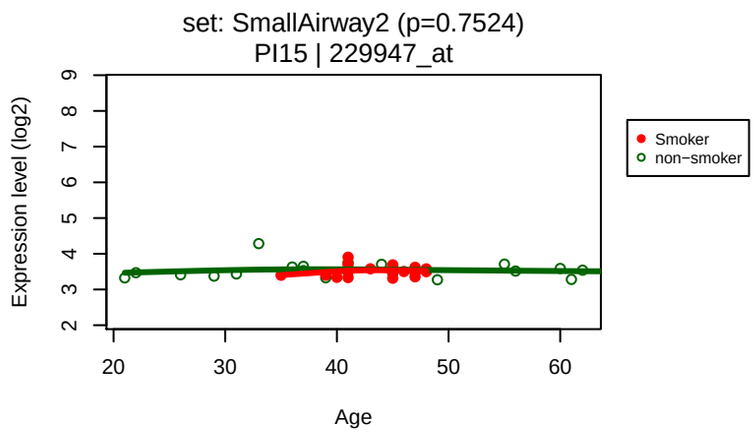
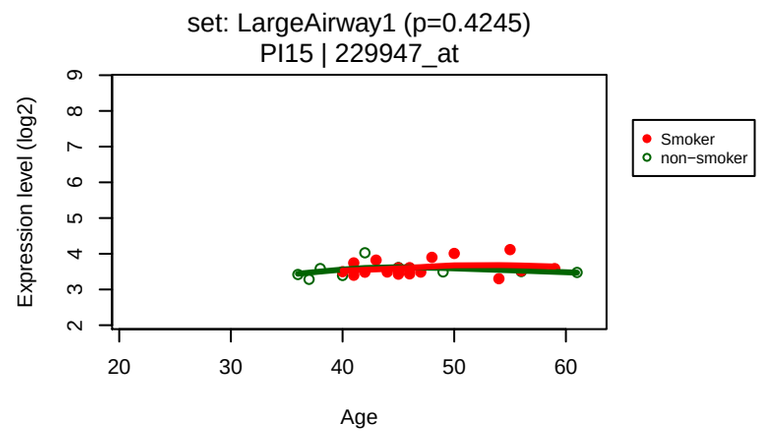

Gene expression of smokers and nonsmokers
gene: PI15 | 229947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4174
p=0.7524
p=0.4082
p=0.4245
p=0.4298
p=0.5611
p=0.7756
p=0.4956
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

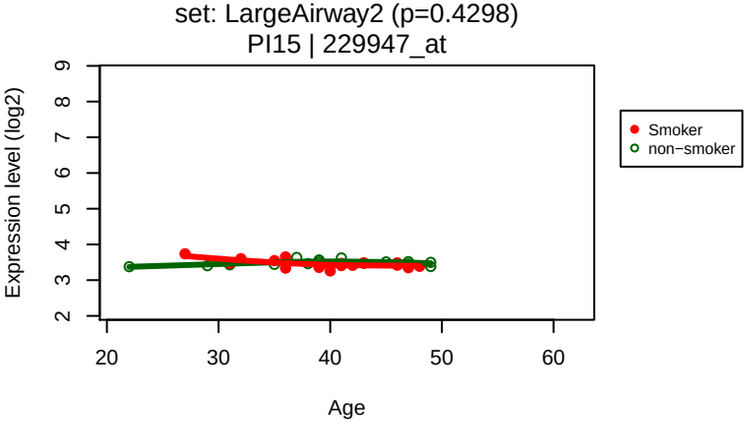

Gene expression ratio (smokers/nonsmokers)
gene: PI15 | 229947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4174
p=0.7524
p=0.4082
p=0.4245
p=0.4298
p=0.5611
p=0.7756
p=0.4956
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

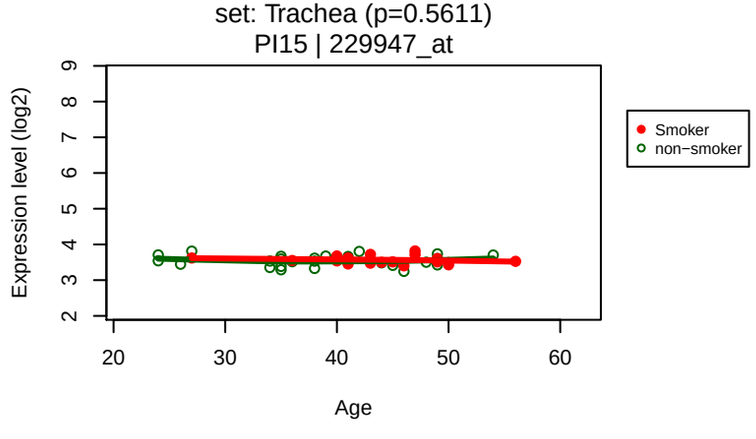

Gene expression ratio in Males
gene: PI15 | 229947_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7904
p=0.1875
p=0.2157
p=0.7662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PI15 | 229947_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9095
p=0.6644
p=0.1274
p=0.425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

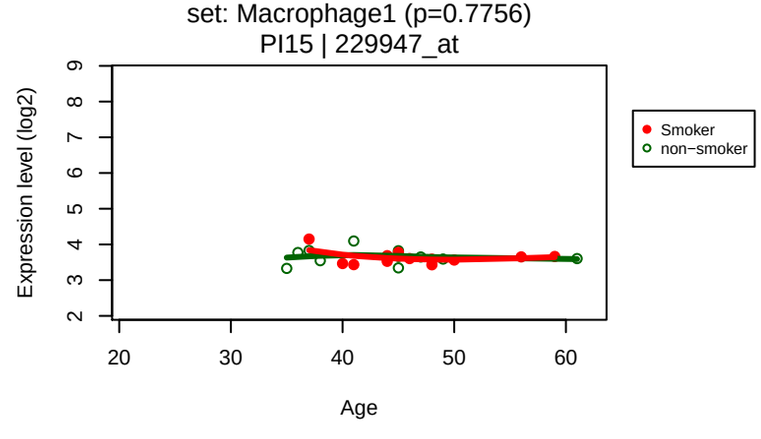

set: SmallAirway1 (p=0.4174)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker

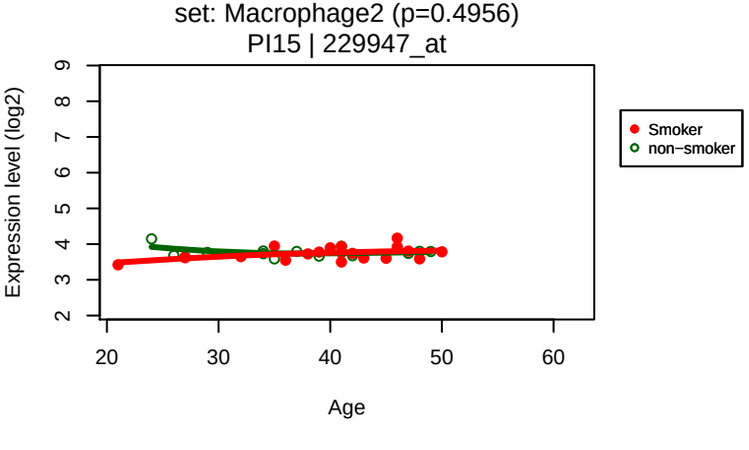

set: SmallAirway2 (p=0.7524)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker

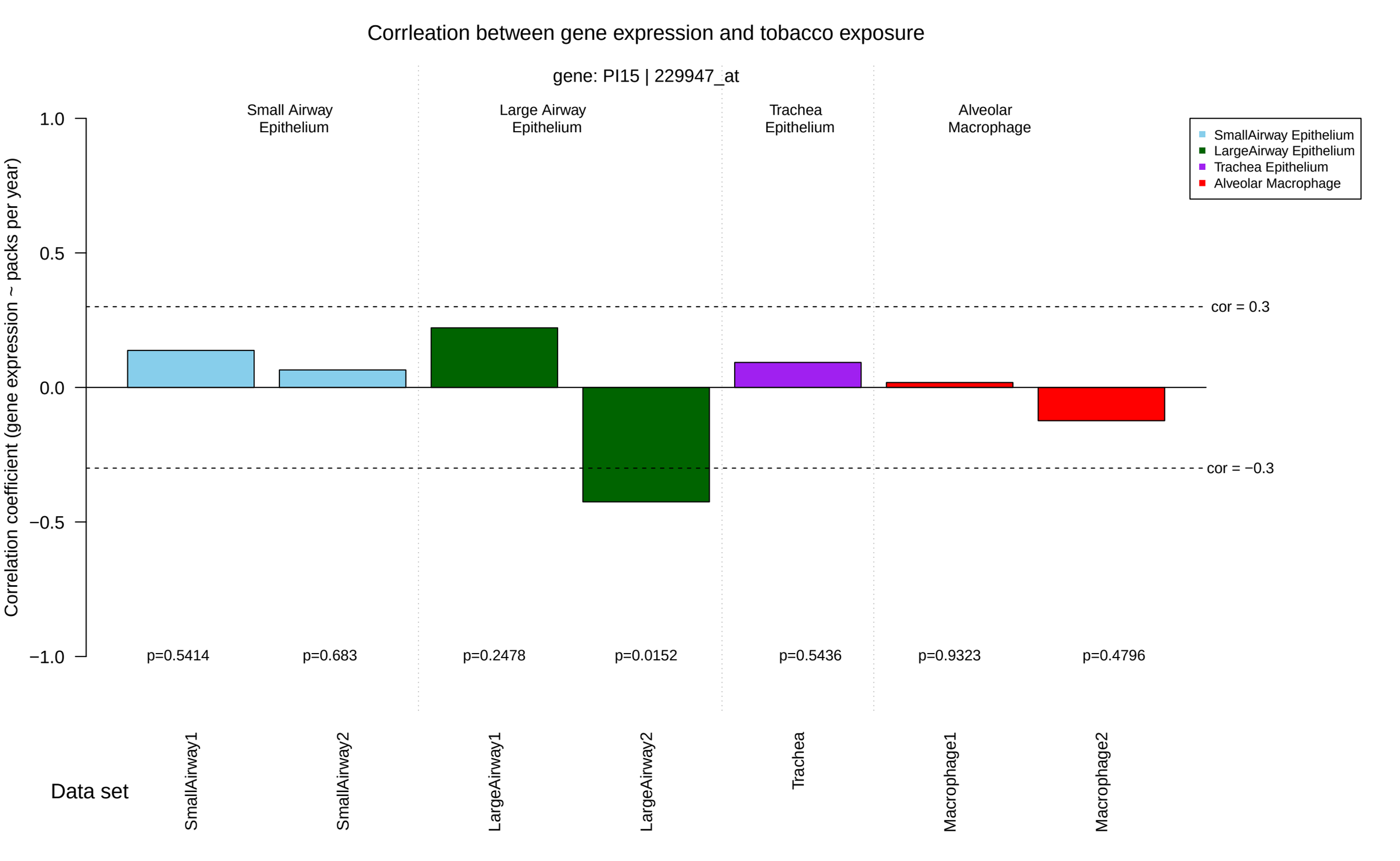

set: LargeAirway1 (p=0.4245)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4298)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5611)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7756)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4956)
PI15 | 229947_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PI15 | 229947_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5414
p=0.683
p=0.2478
p=0.0152
p=0.5436
p=0.9323
p=0.4796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2